

consists of testing the various components of the memory, including testing for functional performance of the address and data lines, and timing tests. The lines are tested using various patterns such as marching ones or pseudo-random values. The values are then read back and compared against expected data for errors. For conducting functional tests on the memory tester, all lines on the memory are tested with sequences such as moving ones, random moving inversion bits, or a random number sequence."

Sanchit Bhatia, digital applications specialist, Agilent Technologies, adds, "DDR is the most popular high-speed memory standard. It is defined by the Joint Electronic Devices Engineering Council (JEDEC) specifications. The JEDEC specifications have strict jitter requirements for the clock and strict signal integrity requirements for all signals. To qualify, the clock must pass a long list of jitter tests and data signals should pass a long list of timing tests. With automated DDR software, jitter measurements can be carried out more efficiently and effectively."

Roy Chestnut, product line manager at Teledyne Lecroy, chimes in, "DDR signalling is a parallel data bus that requires validation of a variety of DDR-specific physical-layer measurements (slew rate, amplitude, overshoot, etc) that are closely defined by the JEDEC standards. These parametric measurements are not standard in oscilloscopes, and it would take users a long time to set up the oscilloscope to make them. Additionally, parallel nature of the DDR bus requires close timing coordination of the various clock, strobe and data signalling. These timing measurements in a variety of test conditions are also automated. Lastly, a variety of fairly standard clock jitter tests are also automated. We have products that focus on command and address timing between the system memory controller and the memory."

Challenges

In the broadest sense, memory testing takes place over the entire life-cycle

The different memory tests

Some of the other tests performed are:

Address lines test. It tests the functionality and connection of address lines. The memory is initialised with all 1's, after which a '0' is written to all the bits of the address. One of the lines is then changed to '1' and using a high-speed digital I/O board (HSDIO), all the lines are read back. The expected value is all 1's. This test is then repeated for all address lines.

Data lines test. In this test, each bit of memory is tested to see whether it can store and return correct values. Once again, all the bits are initialised to '1'. The first bit of an address is changed to '0'. Using the HSDIO board, the bit is acquired and verified as a '0'. This test is then rotated through each bit of the address and repeated for all memory locations.

Self address test. This test writes the value of its own address to each location. Using an HSDIO board, each address is read and checked for values of the data stored. This test is then repeated several times.

Moving inversion one's. In this test, an all 1's moving algorithm is written to the memory and read back. The memory is initialised to all 1's. It is then filled with all 0's from top to bottom verifying each location. The HSDIO board is used to verify the locations. Then the memory is filled with all 1's. The values are read back using the HSDIO board and all the data is verified.

Random moving inversion. Similar to moving inversion one's test, in this test the memory is tested with a moving inversions algorithm using random data. The memory is initialised to all 1's. Random values are written to the memory, from top to bottom, and then verified.

Random number sequence. In this test, random values are written to and read from the memory. This procedure is repeated for several locations.

—Information provided by Abhay Samanth of NI

of a system, beginning with board development or design, moving into production and culminating in post-production stages such as field service. The cycle then repeats when a memory test is performed during the next generation of board design. During each phase in the life cycle, the objectives and goals of memory test differ and the memory test process itself is typically referred to differently, according to the objectives of that particular phase.

Al Crouch, chief technologist at ASSET InterTech, explains in his whitepaper, "During design and new product introduction, testing memory in a timely fashion is particularly critical if the new system is to be delivered promptly to the marketplace. During the initial phase of board bring-up when first prototypes are received, memory tests are performed to identify the root causes of failures or faults in the design of the memory architecture so that these can be quickly corrected prior to the design's release to production. Meanwhile, performance of the memory architecture is characterised to determine whether the design meets or exceeds its performance specifications."

Bhatia says, "The JEDEC specification requires compliance at the hard-to-reach fine ball-grid array (FBGA) package ballout on the dynamic random-access memory (DRAM) chip. Due to difficulty in probing at BGA pins, engineers tend to probe at other locations such as the signal trace or surface-mount components like termination resistors and capacitors. Although this may seem straightforward, signal integrity could be compromised by probing here. First, probing at these locations often causes signal reflection, resulting in non-monotonic edges, overshoot, ringing and other issues. Rather than true signal performance, you see a signal that includes the effect of reflection at components. This can cause errors on slew-rate and setup and hold-time measurements."

"To address this probing challenge and to optimise DDR probing, Agilent has designed specialised tools. One such tool is BGA probe adaptor—a thin fixture that can be attached between the DRAM chip and the circuit board with a compatible footprint on the top and bottom side. The signals at the DRAM ballout are then routed